MODEL A / END-TO-END REPORT

Import-price forecasts

Data, features, training, evaluation and forecast interpretation

2010-2026 | 396 monthly HS rows | 41 features

Separate GBP/kg forecasts for unwrought titanium and other articles. Recorded stress results pass four primary combinations, but calibration timing and mixed median errors require explicit caution.

ASSESSMENT / READ BEFORE USE

Research / demo. Historical gate passes are not strict point-in-time acceptance.

Inside: data sources and preprocessing; features and targets; chronological splits; trained and selected models; validation, disruption and post-disruption comparisons; prediction outputs; graphs; reproducibility and limitations.

This is a formatted PDF edition of the packaged README. The figures are embedded from saved model visualizations. No models, datasets or forecasts were changed. Source documentation: 13 October 2026. PDF generated: 15 October 2026.

PDF bookmarks provide section navigation. Relative file links refer to the ready-to-go package; keep this PDF at its root to retain those references.

Updated 13 October 2026 from this package's saved training code, data, model selection and evaluation records. Documentation only: no retraining or model-selection changes.

Summary And Current Readiness

Model A forecasts the **UK monthly titanium import unit value in GBP/kg**, separately for HS 810820 and 810890. It produces seven quantiles at **H3, H6 and H12**. H3/H6 are the principal acceptance horizons; H12 is reporting-only.

The saved V6 tables record **4/4 primary combinations passing the disruption gates and 3/4 passing post-disruption gates**. Median-price accuracy is mixed: a better uncertainty distribution does not always mean a more accurate central price. These are historical research results, not production approval.

Important calibration-timing limitation: validation includes origin months through January 2022 without restricting all their outcomes to information available before February 2022. For example, January 2022 H6 targets July 2022, and H12 targets January 2023. Nevertheless, build_validation_history() assigns every validation residual an availability date of 31 January 2022. Model selection and the starting calibration pool can therefore use outcomes unavailable at the stress-test start. The recorded passes are **not strictly point-in-time, leakage-free acceptance evidence**. This README exposes the issue; it does not fix or rerun the study.

Data And Sources

Item	What is used
Prepared training file	titanium_model_a_production_v6_titanium_core_corrected_training.csv
Coverage / frequency	January 2010-June 2026; monthly
Size	396 rows, 134 columns; 198 months per HS code
Products	810820: unwrought titanium and powders; 810890: other titanium articles, including bars, sheet and forgings
Excluded	HS 810830 waste and scrap
Trade	UN Comtrade: UK import monetary values, net weight, suppliers and quality information
Currency	Bank of England Database: GBP-USD conversion and FX features
Inflation proxy	ONS price statistics: ERUC/PPI broad non-ferrous import-price movements
Not used	USGS titanium sponge and World Bank aluminium/nickel benchmark features

The CSV is a prepared snapshot, not the original raw downloads. It supports this package's fitting and inference; the complete raw-source collection/preparation archive is not inside V6. Historical absolute paths in manifests remain provenance, not runtime dependencies.

An import unit value is total import value divided by total net weight. For a USD-per-GBP exchange rate, **GBP/kg = USD/kg / USD_per_GBP**. This is not a titanium spot quote: product mix, processing, contract terms, supplier mix and low volumes affect unit values. HS 810820 and 810890 are different product categories, not identical titanium grades.

Preparation, Features And Supervised Targets

Preparation aggregates trade monthly by HS, joins monthly FX/PPI, converts prices to GBP/kg, constructs historical changes and quality diagnostics, and aligns future observations as labels. Quality weights influence fitting. Some early lags and final future targets are naturally unavailable; eligibility filters control fitting, and future actuals are not invented.

There are **41 estimator inputs**, not 134. Exact names are recorded in the model manifest.

Feature names / family	Meaning and usefulness
price_log_change_gbp_1m, _2m, _3m, _6m, _12m	Historical GBP/kg momentum and reversals
price_volatility_6m, price_volatility_12m	Historical variation in price movements
quantity_log_change_1m, _3m, _12m; quantity_volatility_6m, _12m	Volume movement and instability
log_net_weight_kg; volume_ratio_to_prior_12m_median; low_volume_history_flag	Shipment scale and unusually low/high volume; low weights can make prices unstable
supplier_share_usa, _japan, _france, _russia, _china, _other	Supplier/product-mix information
supplier_hhi_weight; top_supplier_share_weight; supplier_concentration_change_3m; top_supplier_share_change_3m; log_partner_count	Concentration, dominant suppliers and diversification
fx_log_change_1m, _3m, _6m, _12m; fx_volatility_6m	Exchange-rate movements affecting GBP prices
eruc_log_change_1m, _3m, _6m, _12m; eruc_volatility_6m	Broad non-ferrous import-price signal, not a titanium-specific benchmark
month_sin; month_cos	Cyclic calendar seasonality
price_shock_history_flag; world_weight_estimated_by_un_flag; eu_collection_change_flag	Observed-history/data-quality/collection-regime controls

Feature names abbreviated with suffixes above share the displayed prefix. Dates, future target columns and availability controls are not estimator features. Derived inputs are calculations from the source history, not random synthetic observations. Feature importance shows statistical association, not causation.

The supervised regression target is **ln(price[t+h] / price[t])**. Convert output q to price with **price[t] * exp(q)**. This is the monthly price h months ahead, not the average of all intervening months. H6/H12 are direct fitted forecasts, not repeated H3 predictions.

Horizon	Fitted target column	Corresponding actual price
H3	target_log_change_gbp_per_kg_h3	target_unit_value_gbp_per_kg_h3
H6	target_log_change_gbp_per_kg_h6	target_unit_value_gbp_per_kg_h6
H12	target_log_change_gbp_per_kg_h12	target_unit_value_gbp_per_kg_h12

Target-month and assumed target-availability columns supply alignment. Retained USD labels are not the GBP target fitted in V6.

Chronological Splits

This is an expanding-history study, not a random split. Each validation block fits eligible history at its first assumed availability cutoff. Test fits refresh monthly and may use earlier test outcomes after they become assumed available; the selected algorithm remains frozen.

Stage	Forecast-origin months	Notes
Initial history	2010-2016 before the first validation block	Lag completeness, reliability and target availability reduce actual fit rows
Validation block 1	January-December 2017	Expanding history
Validation block 2	January-December 2018	Expanding history
Validation block 3	January-December 2019	Expanding history
Validation block 4	February 2020-January 2021	January 2020 is not in the configured blocks
Validation block 5	February 2021-January 2022	60 forecasts per HS/horizon pooled for selection/calibration
Disruption stress	February 2022-December 2023	23 origins per HS/horizon; war-era stress window
Post-disruption	January 2024-December 2025	24 origins; additional reused historical evaluation
Latest saved forecast	June 2026 reference observation	Final bundles use eligible completed labels; actual future prices are unavailable in this snapshot

Origin dates and target dates are different. Stress H3 targets run May 2022-March 2024, H6 August 2022-June 2024 and H12 February 2023-December 2024. Post H3/H6 targets end March/June 2026. Post H12 targets extend through December 2026, so only 18 of 24 outcomes are observed.

These are this V6 implementation's periods, not the earlier proposed generic split for all three models. Assumed release dates and current revised historical snapshots do not establish verified point-in-time availability.

Algorithms And Saved Selection

Candidate	Learning / uncertainty method
Linear quantile	Regularized, quality-weighted quantile regressions
Gradient Boosting quantiles	Nonlinear quantile-loss regressors
Random Forest distribution	Weighted tree ensemble; quantiles of individual-tree predictions
Elastic Net residual	Standardized regularized central regression plus residual quantiles
Huber residual	Robust central regression plus residual quantiles
Adaptive no-change	No learned feature-to-price mapping; no-change distribution with volatility scaling
Fixed no-change baseline	Separate benchmark distribution and calibration history

The adaptive distribution scales around its median using current 6-month volatility relative to its training reference, with the implemented scale clipped to 0.80-1.80. Quantiles are non-crossing. Calibration expands P05/P95; rolling evaluation pools up to 36 available residuals and prefers the same volatility regime when sufficient history exists.

Selection minimizes calibrated validation MIS among gate-eligible candidates; raw pinball loss breaks ties. The same pooled validation residuals calibrate the validation intervals, so their coverage is not an independent calibration holdout.

HS	H3	H6	H12 / reporting-only
810820	Linear quantile	Adaptive no-change	Random Forest distribution
810890	Elastic Net residual	Linear quantile	Elastic Net residual

810820 saved bundle and 810890 saved bundle each contain three fitted horizon entries, preprocessing and calibration metadata. They are not single multi-output estimators.

Evaluation Metrics

Metric	Meaning
MAE / RMSE	P50 absolute error / root mean squared error in GBP/kg; RMSE penalizes large misses more
MAPE / sMAPE	Mean percentage deviation / symmetric percentage deviation; neither is classification accuracy
Pinball loss	Average quantile loss over seven outputs; lower is better
P05-P95 coverage	Observed fraction within the nominal 90% interval; not guaranteed future coverage
Mean Interval Score (MIS)	Width plus miss penalties at alpha 0.10, here in log-change units
MIS / baseline	Below 1 means improved interval score versus the separately calibrated baseline
Recovery	Six-origin rolling coverage by August 2022 using eventual outcomes, not six outcomes already published then

Recorded H3/H6 acceptance requires coverage >=90%, better MIS, complete outputs and that recovery check. H12 is outside the pass count. A baseline's false pass flag is expected because it cannot beat itself. Recovery here is not equivalent to Model C's completed-outcome recovery definition.

Selected Validation Results

HS / horizon	Coverage	MIS	MIS / baseline
810820 H3	93.33%	0.8540	0.9701
810820 H6	93.33%	0.9442	0.9780
810820 H12	93.33%	0.8351	0.7739
810890 H3	93.33%	2.2366	0.8692
810890 H6	93.33%	2.1693	0.7962
810890 H12	93.33%	2.3062	0.7998

Full algorithm comparison. Identical coverage arises from the same-size calibrated pools; it is not independent confirmation of six future success rates.

Disruption And Post-Disruption Results

Period	HS / horizon	Coverage	MIS	MIS / baseline	Pinball loss	Recorded gate
Stress	810820 H3	100.00%	0.5912	0.5722	0.0287	Pass
Stress	810820 H6	100.00%	0.8546	0.7991	0.0376	Pass
Stress	810890 H3	95.65%	2.7357	0.9671	0.0673	Pass
Stress	810890 H6	100.00%	2.0481	0.7769	0.0642	Pass
Post	810820 H3	95.83%	0.8307	0.8275	0.0386	Pass
Post	810820 H6	100.00%	0.8212	0.7395	0.0279	Pass
Post	810890 H3	100.00%	1.8680	1.0371	0.0386	Fail MIS
Post	810890 H6	100.00%	1.8069	0.8674	0.0515	Pass

Stress metrics and post metrics. Primary rows have all 23/24 outputs. Post H12 has 75% recorded coverage and infinite MIS because six unknown outcomes are counted as misses; it is incomplete outcome evidence, not proof those unknown forecasts were wrong.

Median Errors Versus Baseline

Recalculated from saved prediction rows with positive observed actual prices and finite P50. MAE/RMSE are GBP/kg. Model MAPE is not the selection objective.

Period	HS / horizon	MAE	RMSE	Model MAPE	Baseline MAPE	Model sMAPE
Stress	810820 H3	0.8838	1.1217	8.12%	10.32%	8.43%
Stress	810820 H6	1.3980	1.6942	12.26%	12.26%	13.20%
Stress	810890 H3	7.6000	10.5162	20.02%	19.85%	16.88%
Stress	810890 H6	8.2450	10.5279	20.75%	18.53%	17.99%
Post	810820 H3	1.3313	2.1718	12.11%	9.10%	10.57%
Post	810820 H6	0.9087	1.1097	8.06%	8.06%	7.87%
Post	810890 H3	4.1186	5.5263	9.33%	10.25%	9.40%
Post	810890 H6	6.5035	7.4881	14.28%	7.93%	14.45%

Stress actual/predicted records and post records. Compare **actual_target_price_gbp_per_kg** with **p50_price_gbp_per_kg**. actual_origin_price_gbp_per_kg is the starting price, not ground truth for the future month. Signed error = P50 - actual; absolute percentage error = 100 * abs(P50 - actual) / actual.

810820 H3 improves stress median MAPE but not post MAPE. Adaptive H6 has the same median as baseline; its improvement is interval calibration. 810890 H6 has better interval scores but worse median MAPE than baseline. High coverage alone can reflect wide intervals; MIS penalizes width, and timing limitations still apply.

Latest Saved Forecasts

Reference observation: **June 2026**, not the computer's current month. These are unseen target values relative to the prepared snapshot; a past calendar month is not automatically an observed actual in this dataset.

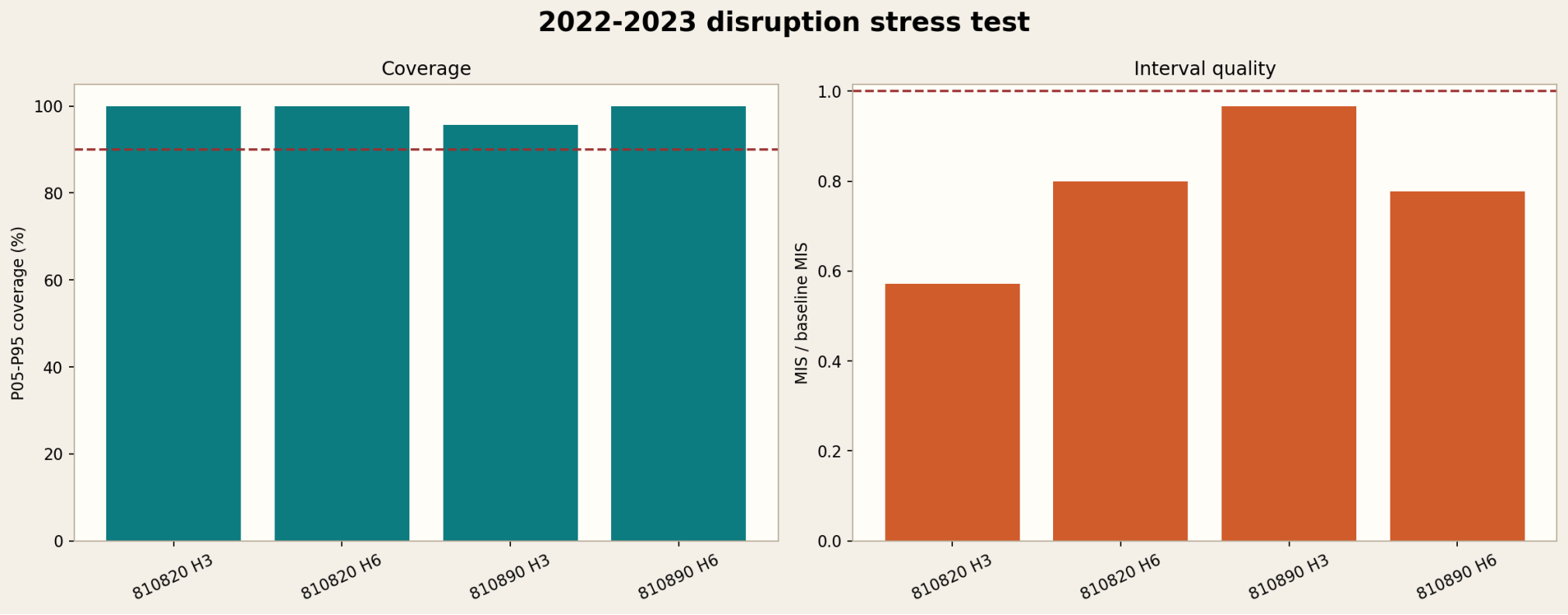
HS	Horizon / target	P05 GBP/kg	P50 GBP/kg	P95 GBP/kg
810820	H3 / September 2026	8.3006	12.6599	15.7657
810820	H6 / December 2026	8.6626	10.9202	14.4837
810820	H12 / June 2027	8.4861	10.7770	13.1116
810890	H3 / September 2026	31.2629	50.8343	82.2818
810890	H6 / December 2026	29.2205	50.7638	83.9316
810890	H12 / June 2027	25.5961	53.0612	103.1928

Full forecast CSV, response JSON, manifest. Outputs include seven quantiles, HS/product, horizon, origin/target month, selected model, currency/unit and research status.

810890's high P95 values describe large upper-tail uncertainty, not its expected central price. Quantiles must be ordered within a horizon, but P95 or P50 need not increase with horizon. Future accuracy requires actual outcomes.

Figure 1. Stress coverage and MIS

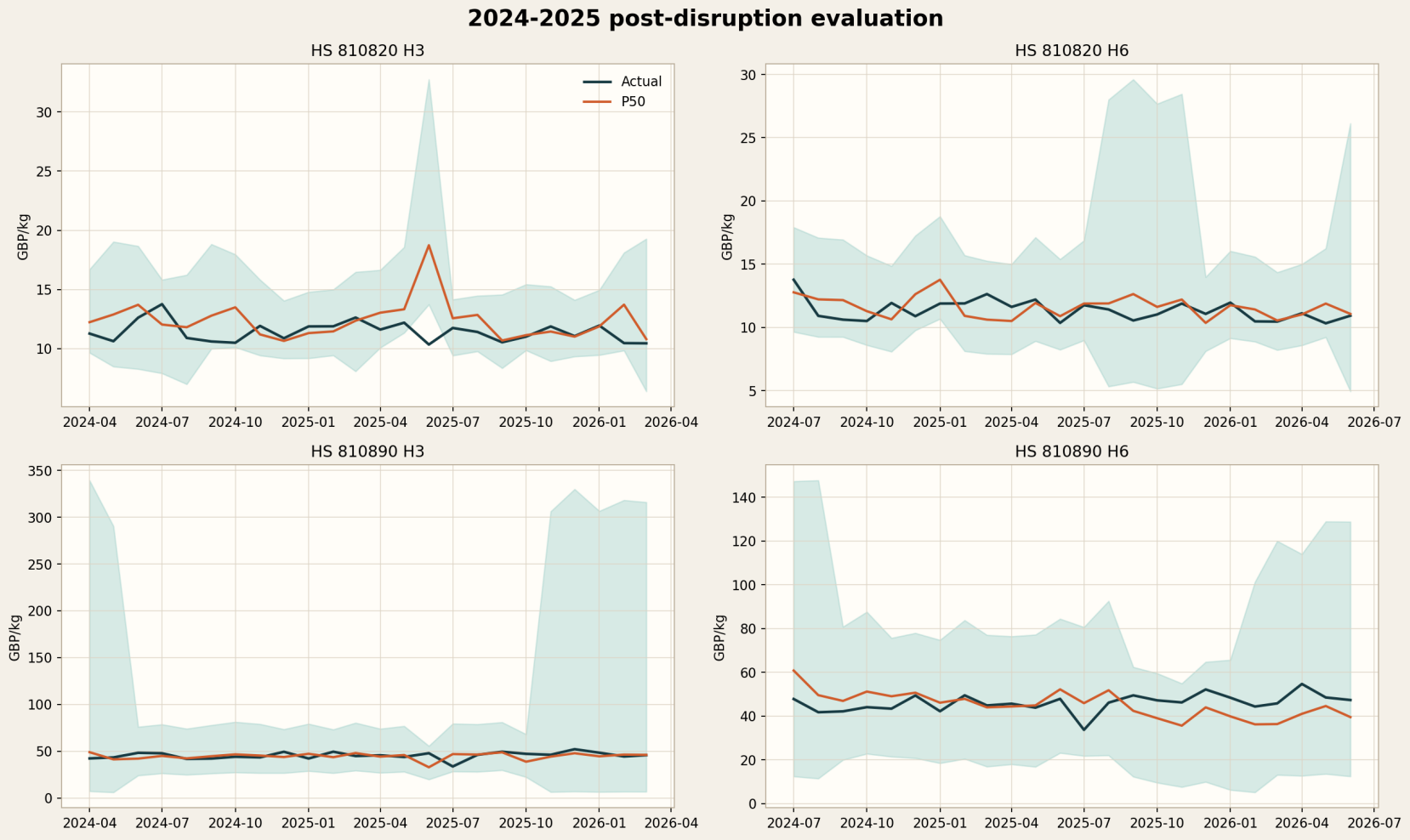
Saved visualization / Graphs And Dashboards



Source: visualizations/stress_2022_2023_coverage_and_mis.png

Figure 2. Post-disruption actual price, predicted price and intervals

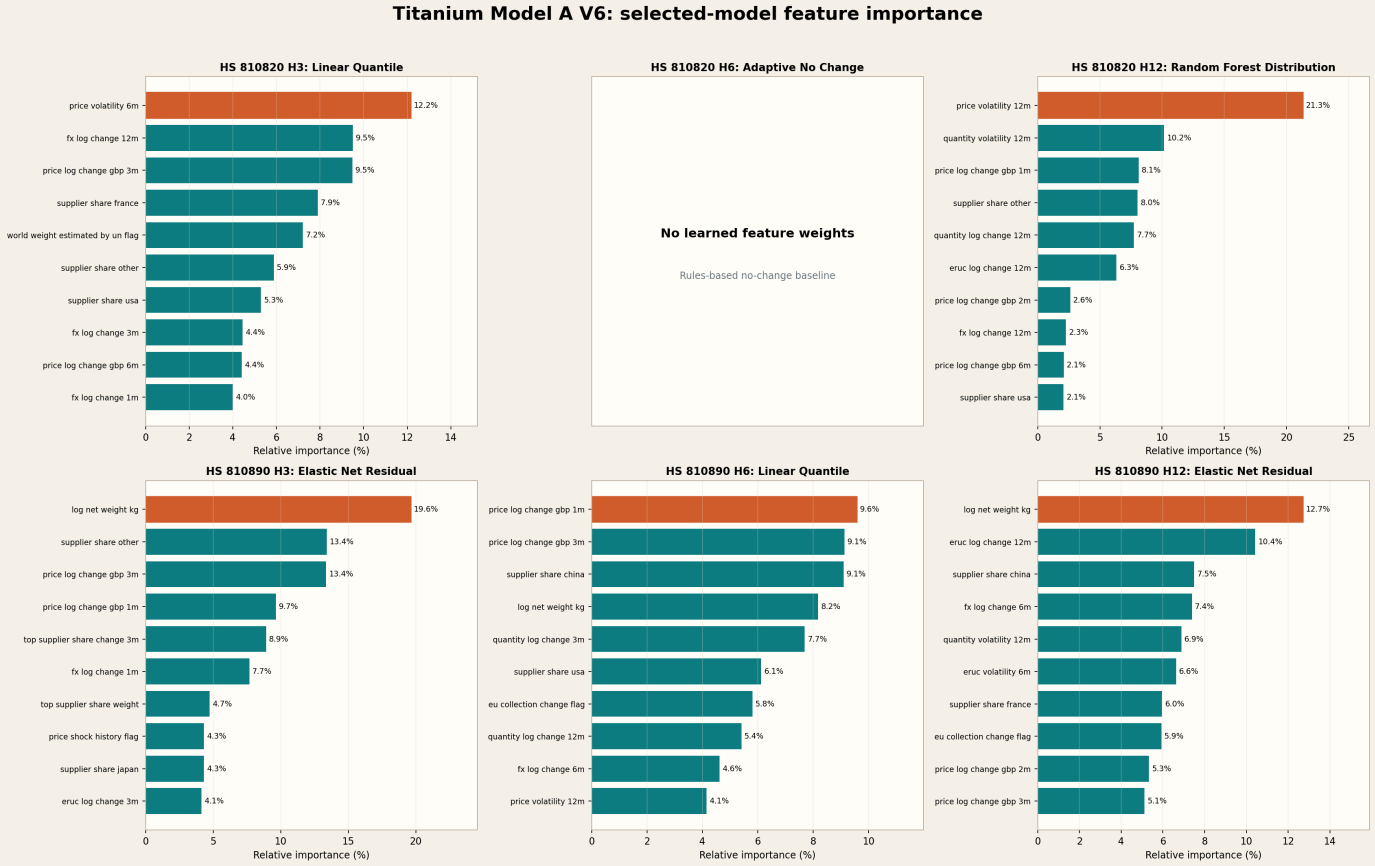
Saved visualization / Graphs And Dashboards



Source: visualizations/post_2024_2025_actual_vs_forecast.png

Figure 3. Selected-model feature influence

Saved visualization / Graphs And Dashboards



Source: visualizations/selected_model_feature_importance.png

Post coverage/MIS | Reliability over time | Performance dashboard | V6 dashboard.

GitHub displays the PNGs. Download HTML pages and open them locally for interactive content; GitHub's file viewer does not run the dashboards.

Folder Map And Reproduction

Location	Contents
data/	Package-local prepared training snapshot
code/ and code/engines/	V6 training, local inference, engine dependencies, dashboard/PDF utilities
models/	Two HS bundles with three horizons each
results/	Validation, stress, post and latest records
visualizations/	PNG/HTML reports
monitoring/	Feature drift, historical forecasts and data-quality diagnostics
documentation/	Model card and PDF records

From this V6 directory: `python -m pip install -r requirements.txt`, then `python code/saved_model_inference.py`. Verification reproduces **42 latest quantile-price values**, without training. The combined root's VERIFY_PACKAGE.cmd checks all models and the UI.

RUN_TRAINING_V6.cmd retrains and writes artifacts. Do not run it merely to view saved results or overwrite the chosen evidence without preserving a copy/new run directory. Packaging changed local import paths only; saved models are unchanged. Load only trusted joblib files.

Limitations And Next Work

This is a caveated research/demo package, not an unqualified production pricing model. Correct availability-purged selection/calibration, evaluate realistic completed-outcome recovery, obtain verified release dates/vintages, and validate on an untouched prospective holdout before production use. Earlier experiments inspected these historical periods. ERUC is a broad proxy and product-mix effects remain substantial.

Saved-value reproduction confirms implementation consistency, not future accuracy. A's MIS, B's Brier and C's index error are different measures and should not be numerically ranked against each other. Read the data reuse review before public redistribution. Pinned versions match the checked environment; clean-environment installation has not been tested.